JOYCE LUO

585-203-7590 / joyceluo@mit.edu / LinkedIn / Website

EDUCATION

Massachusetts Institute of Technology, Cambridge, MA ***Expected May 2027***
PhD in Operations Research
Advisor: Prof. Georgia Perakis

Princeton University, Princeton, NJ ***May 2022***
BSE in Operations Research & Financial Engineering (ORFE)
Minors: Statistics & Machine Learning, Computer Science
Thesis Advisor: Prof. Bartolomeo Stellato

RESEARCH INTERESTS

- Healthcare operations management: telehealth platforms, hospital operations, public health, equity in healthcare systems, human-AI interaction in healthcare, resource allocation, intervention design
- Developing data-driven methodologies integrating optimization, machine learning, causal inference, natural language processing (large language models, deep learning)

PUBLICATIONS

1. Canellas, M., **Luo, J.,** Pachamanova, D., Perakis, G.* (2026). **Fair online hospital diagnostic service scheduling: Helping both patients and providers.** *Production & Operations Management*, accepted and available online.

2. **Luo, J.,** Stellato, B. (2024). **Frontiers in Operations: Equitable Data-Driven Facility Location and Resource Allocation to Fight the Opioid Epidemic.** *Manufacturing & Service Operations Management*, 26(4), 1229-1244.

3. **Luo, J.**, Chen, L., Lu, X., Yuan, J., Xie, Z., Li, D. (2021). **Analysis of potential associations of JUUL flavours with health symptoms based on user-generated data from Reddit.** *Tobacco Control*, 30(5), 534-541.

4. Lu, X., Chen, L., Yuan, J., **Luo, J.**, Luo, J., Xie, Z., Li, D. (2020). **User Perceptions of Different Electronic Cigarette Flavors on Social Media: Observational Study.** *Journal of Medical Internet Research*, 22(6), e17280.

5. Chen, L., Lu, X., Yuan, J., **Luo, J.**, Luo, J., Xie, Z., Li, D. (2020). **A Social Media Study on the Associations of Flavored Electronic Cigarettes With Health Symptoms: Observational Study.** *Journal of Medical Internet Research*, 22(6), e17496.

6. **Luo, J.,** Dehghani, M., Vahdat, V., Alagoz, O., Estes, C., Tyson, C., Ebner, D.W., Limburg, P.J. **Adherence Modeling in Colorectal Cancer Screening Through the Longitudinal Adherence Trajectory (LAT) Framework.** Major Revision at *Medical Decision Making*.

7. **Luo, J.,** Pachamanova, D., Perakis, G.* **Discovering Latent Treatments in Unstructured Observational Data: A Framework for Causal Effect Estimation and Personalized Intervention Design.** Submitted to *Management Science*.

8. **Luo, J.,** Pachamanova, D., Perakis G., Zhang K.* **Optimization-Based Active Learning for Multi-Label Text Classification.** Working paper.

(* indicates authors listed alphabetically)

PRESENTATIONS

Conference Presentations

A Causal Framework for Latent Treatments: Evaluating Supporter Communications in Digital Mental Health Services.

- 2026 INFORMS Healthcare Conference, July 2026
- 2026 MSOM Conference, July 2026
- 2026 POMS Conference, May 2026

Fair Online Hospital Diagnostic Service Scheduling: Helping Both Patients and Providers.

- 2026 POMS Conference, May 2026
- 2025 INFORMS Annual Meeting, October 2025
- 2025 MSOM Conference, June 2025

Equitable Data-Driven Facility Location and Resource Allocation to Fight the Opioid Epidemic.

- 2025 MSOM Conference, June 2025

Addressing Healthcare Provider Burnout: Fair Online Hospital Diagnostic Service Scheduling.

- 2025 POMS Conference, May 2025
- 2024 INFORMS Annual Meeting, October 2024

To Open or Not to Open: Efficient Scheduling and Capacity Management for Diagnostic Services.

- 2023 INFORMS Annual Meeting, October 2023

Equitable Data-Driven Resource Allocation to Fight the Opioid Epidemic: A Mixed-Integer Optimization Approach.

- Princeton Research Day, May 2022
- Princeton Center for Statistics & Machine Learning Undergraduate Research Poster Session, May 2022
- Top ORFE Theses Department Showcase, May 2022

Analysis of potential associations of JUUL flavours with health symptoms based on user-generated data from Reddit.

- WNY CRoFT Virtual Annual Meeting, September 2020

Workshops

- Invited to 2026 Cornell ORIE Young Researchers Workshop, October 2026
- Invited to 2026 Tippie FutureBAProf Workshop, August 2026
- Invited to 2025 MIT Path of Professorship Workshop, November 2025
- Presented at 2025 MIT Sloan Health Systems Initiative Lab Workshop, October 2025
- Presented at 2024 MIT Sloan Health Systems Initiative Lab Workshop, October 2024

SELECTED AWARDS & HONORS

- **National Science Foundation Graduate Research Fellowship** (2022-Present)
- **Sigma Xi Book Award for Outstanding Research**, Princeton ORFE Department (2022)
- **Undergraduate Research Poster Session Winner**, Princeton Center for Statistics & Machine Learning (2022)
- **First Place Undergraduate Poster Presentation**, WNY CRoFT Virtual Annual Meeting (2020)

TEACHING EXPERIENCE

Teaching Assistant, 15.730 Data, Models, and Decisions ***January 2026-June 2026***
Rating: 6.4/7.0

- MIT Sloan Executive MBA core curriculum course with 120+ EMBA students

- Organized and taught recitations, conducted office hours, graded student assignments, and developed course material

Teaching Assistant, 15.774/15.780 Analytics of Operations Management ***September 2025-December 2025***
Rating: 6.64/7.0
- MIT Sloan MBA and undergraduate elective course with 70+ MBA students and 30+ undergraduates
- Organized and taught recitations, conducted weekly office hours, graded student assignments, and developed course material

Lecturer, 15.S60 Computing in Optimization and Statistics ***January 2025***
- Taught 3-hour lecture on machine learning in Python for PhD students

Teaching Assistant, 15.730 Data, Models, and Decisions ***January 2024-May 2024***
Rating: 6.7/7.0
- MIT Sloan Executive MBA core curriculum course with 120+ EMBA students
- Organized and taught recitations, conducted office hours, graded student assignments, and developed course material

Undergraduate Course Assistant, ORF 307 – Optimization ***January 2022-May 2022***
- Provided detailed feedback on student assignments and assisted with course organization

Undergraduate Course Assistant, ORF 245 – Fundamentals of Statistics ***January 2021-May 2021***
- Provided detailed feedback and assisted students with their assignments to contribute to their learning of the course material

Tutor, McGraw Center for Teaching and Learning ***September 2019-December 2021***
- Tutored students in engineering calculus and physics courses approximately 3 hours per week
- Facilitated group discussion and understanding about complicated topics in a study hall setting

RESEARCH EXPERIENCE

Doctoral Research Assistant, Massachusetts Institute of Technology, Cambridge, MA ***September 2022-Present***
- Current projects related to improving patient wait times and provider resource allocation for hospital diagnostic services through a novel online algorithm; evaluating support messaging for mental health patients through a causal machine learning, optimization, and large language model-based framework

Undergraduate Senior Thesis, Princeton University, Princeton, NJ ***June 2021-April 2022***
- Formulated an optimization problem that incorporates a predictive ordinary differential equation-based model of the US opioid epidemic and socioeconomic equity considerations to recommend opioid treatment facility locations and treatment budget allocations across the US

Undergraduate Researcher, Princeton University BEEHIVE Group, Princeton, NJ ***December 2020-February 2022***
- Worked on an offline reinforcement learning algorithm for continuous action spaces that recommends anticoagulant dosing policies for ICU patients, using electronic health records (EHR) data

Research Intern, University of Rochester CTSI, Rochester, NY ***May 2019-June 2020***
- Conducted data mining, data modeling, and statistical analysis using Python and SAS on social media data from Reddit to make conclusions about potential associations between different JUUL (e-cigarette) flavors and health problems

PROFESSIONAL SERVICE

Ad-Hoc Reviewer
- MSOM Conference Extended Abstracts (2025, 2026)

- *IEEE Transactions on Automation Science and Engineering*

Conference Volunteer, 2026 MSOM Conference ***July 2026***

Co-Chair, Conference in Celebration of Georgia Perakis ***March 2026***

Seminar Series Coordinator, MIT ORC Spring Seminar Series ***January 2025-May 2025***

Vice President, MIT INFORMS Student Chapter ***December 2022-December 2023***

Co-President, Princeton Data Science (PDS) ***November 2019-April 2022***

Board Member - Girl Scout Day Chair, Princeton Society of Women Engineers ***September 2018-January 2020***

PROFESSIONAL EXPERIENCE

Health Economics & Outcomes Research Intern, Exact Sciences, Cambridge, MA ***June 2025-August 2025***

- Developed a predictive model for capturing longitudinal patient adherence to various colorectal cancer (CRC) screening modalities
- Created a calibration procedure combining a multi-start optimization algorithm with simulation to fit the predictive model to population-level CRC screening adherence data
- Integrated the calibrated longitudinal adherence models with the company's CRC microsimulation pipeline used for various cost-effectiveness analyses

Program Manager Intern, Microsoft, Redmond, WA (Remote) ***June 2021-August 2021***

- Designed and documented a methodology, which consists of data collection and statistical analysis, that measures the entropy (randomness) of random number generation systems
- Created a justification document with results from the methodology, ensuring that Microsoft's random number generation infrastructure meets government standards and can be used by commercial and government partners
- Analyzed data and conducted energy calculations to estimate the carbon footprint of government-mandated self-tests in Windows 10 and Windows Server, and presented findings in a research report

Explore Intern, Microsoft, Redmond, WA (Remote) ***June 2020-September 2020***

- Developed and automated the creation of a virtual machine that provides enhanced security for Azure customers
- Defined the customer experience by creating product specs and engaging with customers
- Created impactful data dashboards that are used by product managers to gauge product usage

SKILLS & CERTIFICATIONS

Computer Skills: Python, Julia, R, Java, LaTeX, PowerShell, Stata, MATLAB, AMPL, SAS, SQL, C#, C++
Certifications: MIT Path of Professorship Certificate
Additional Languages: Chinese (Fluent)